

## Original Report

# The association between event learning and continuous quality improvement programs and culture of patient safety



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### Abstract

**Purpose:** To present our approach and results from our quality and safety program and to report their possible impact on our culture of patient safety.

**Methods and materials:** We created an event learning system (termed a “good catch” program) and encouraged staff to report any quality or safety concerns in real time. Events were analyzed to assess the utility of safety barriers. A formal continuous quality improvement program was created to address these reported events and make improvements. Data on perceptions of the culture of patient safety were collected using the Agency for Health Care Research and Quality survey administered before, during, and after the initiatives.

**Results:** Of 560 good catches reported, 367 could be ascribed to a specific step on our process map. The calculated utility of safety barriers was highest for those embedded into the pretreatment quality assurance checks performed by physicists and dosimetrists (utility score 0.53; 93 of 174) and routine checks done by therapists on the initial day of therapy. Therapists and physicists reported the highest number of good catches (24% each). Sixty-four percent of events were caused by performance issues (eg, not following standardized processes, including suboptimal communications). Of 31 initiated formal improvement events, 26 were successfully implemented and sustained, 4 were discontinued, and 1 was not implemented. Most of the continuous quality improvement program was conducted by nurses (14) and therapists (7). Percentages of positive responses in the patient safety culture survey appear to have increased on all dimensions ( $p < .05$ ).

**Conclusions:** Results suggest that event learning and continuous quality improvement programs can be successfully implemented and that there are contemporaneous improvements in the culture of safety.

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### Introduction

Available data suggest that potential quality/safety events occur during the course of treatment in ~1% to 3% of patients, but the vast majority of these are not clinically

relevant.<sup>1</sup> This compares unfavorably with highly reliable industries such as commercial aviation (~1 death in 4.7 million passenger flights) or other areas of medicine such as anesthesiology (~1 death in 200,000 procedures).<sup>2</sup> However, these comparisons might not be totally fair because the reporting thresholds are different. If in aviation we were to count faulty takeoffs, landings, or unplanned returns to the airport, and if in anesthesiology we reported intubation failures or ventilator equipment/tube malfunctions, aviation and anesthesiology might not appear as favorable.

Nevertheless, the relatively high rate of any type of event within radiation oncology is a cause for concern because it suggests inherent shortcomings of our current systems. A series of articles in *The New York Times*<sup>3-6</sup> highlighted major quality and safety concerns. In response to these articles, the American Society for Radiation Oncology, American Association of Physicists in Medicine, The Society of Radiation Oncology Administrators, the American Association of Medical Dosimetrists, and the American Society of Radiologic Technologists all issued leadership statements regarding patient safety; moreover, the presidents of American Society for Radiation Oncology and the American Association of Physicists in Medicine were called to a congressional hearing in Washington, DC.<sup>7-11</sup> The statements released by these societies, combined with the congressional hearing, led to radiation therapy professional societies taking a greater leadership role in instituting change and elevating the importance of proactive quality and safety efforts.

A major theme in many of the national statements is the need to better foster a culture of safety. Other high reliability and value creation industries, including some health care organizations, have successfully promoted a culture of safety (at least in part) by developing and fostering event reporting and analysis mechanisms and engaging in continuous quality improvement (CQI) efforts.<sup>12-17</sup>

For example, Virginia Mason Hospital, one of the nation's leading hospitals in quality and safety, requires all staff to report unsafe conditions and (if needed) initiate (or at least participate) in associated CQI initiatives. From the program's beginning in 2002, more than 20,000 reports have been made and helped to drive successful improvement activities.<sup>17</sup>

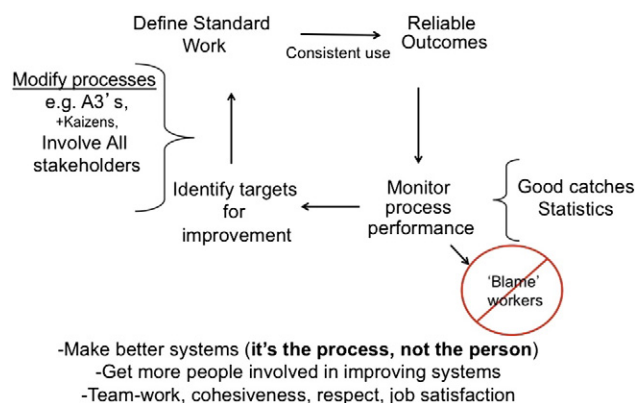
Toward this aim, over the past six years, we have created a multifaceted quality and safety improvement program in our department. This includes a formal system to define standard work, a systematic approach to report and analyze events, and the use of pooled data to help us guide CQI activities. These components are mutually reinforcing (Fig 1). The entire structure is promoted by enthusiastic leadership and an extensive support infrastructure (eg, educational programs, software tools, administrative assistance). We aim to involve all employees in these activities.

To share our experience and to assess how well our quality and safety improvement program is functioning, in this article we will do the following:

1. Summarize a systematic analysis of our reported events (including where in our processes events were initiated and the effectiveness of existing safety barriers);
2. Summarize our formal improvement initiatives (including how well our improvement initiatives do-or do not-address steps in our processes where events were reported); and
3. Assess the possible impact of our quality and safety program on our department's culture of patient safety.

## Methods and materials

Our quality and safety improvement program is run and supported by a multidisciplinary quality and safety committee led by a physician. The committee meets weekly to analyze the reported events and to initiate and monitor progress on CQI efforts.



**Figure 1** High-level summary of the processes and infrastructure to support our quality and safety program. Our systems are never complete as we constantly adjust and improve them in order to solve problems and innovate. Because each element is not as useful in isolation, the period of construction of the entire system can be frustrating.

## Event learning program

Our Web-based event learning program (termed the “good catch” program to provide a positive connotation and promote a “no blame” message to staff reporting events) is available for all staff to easily report any quality or safety concern in real time (ie, something that reached the patient, a near miss, or an unsafe condition). In addition to the weekly meeting, the key insights from event analysis are reviewed in the monthly quality assurance (QA) meeting with the whole department. To promote and encourage participation in this program, at this meeting we publicly recognize an employee who had the most seminal good catch for that month. Staff photographs and descriptions of good catches are posted on bulletin boards in several locations within the department. Recognized employees also receive a \$30 voucher to use at the hospital coffee shop or cafeteria and sign the department basketball, which is prominently displayed in our departmental trophy case. In other words, we publicly celebrate individuals who raise meaningful concerns about quality and safety.

We analyzed 560 events reported between June 2012 and May 2014. For each event, we mapped where it was initiated and caught (or not caught) using our process map. Furthermore, root causes (or key contributing factors) are identified and categorized using a high-level taxonomy (eg, “performance” issues [not following or understanding policies and procedures, sending suboptimal communications], lack of process standardization, technological/environmental factors [difficulties with information technology, suboptimal workload, etc.]).<sup>18</sup> The number of events generated at each step and propagated from the prior steps and events caught and corrected at various steps in our workflows was used to estimate the utility of safety barriers at each process step, using the number of events caught by the safety barriers divided by the number of events that were presented to the safety barriers at the respective process step.

We also analyzed reporting efforts by professional groups (physicians, physicists, dosimetrists, therapists, nurses, and administration/clerical staff).

## Formal CQI program

Many of our improvement activities are performed in the context of an A3 (a problem-solving tool that follows the formal Plan-Do-Study-Act cycle). A3s were invented/popularized by Toyota, and the name is derived from the paper size used for the report—the metric equivalent to 11 × 17-inch paper.<sup>19-23</sup> The A3 program is run by a program manager, an industrial engineer specializing in problem-solving, process improvement, and change management. Managing the A3 program includes training, ongoing coaching, an approval process, an implementation process including “rapid” and structured improvement events (also called Kaizen events), 30-/60-/90-day sustainability checks, visual management, and rewards and/or recognition. Specifically, staff are recognized and rewarded

both individually and collectively for participating in the A3 program. New A3s are highlighted at the monthly departmental QA meeting and posted on the visual management board. For each A3 implemented, the area in which the A3 owner belongs (eg, nursing, physics, administration) receives \$100 in its “bank” that can be used for additional quality and safety improvements in their areas. For example, the nurses implemented several A3s, earned \$800, and installed a monitor in the nurse’s work room to display the queue of patients in the lobby so they could better monitor their patient flow and minimize patient wait times. We analyzed 33 A3s implemented between June 2012 and May 2014 in terms of which steps of our processes they address to reveal how well those A3s were linked to reported good catches and their apparent impact on our clinical operations. We also analyzed A3 efforts by professional groups (ie, physicians, physicists, dosimetrists, therapists, nurses, and administration/clerical staff).

## Culture of patient safety

The Agency for Healthcare Research and Quality (AHRQ) defines the safety culture of an organization “as the product of individual and group values, attitudes, perceptions, competencies, and patterns of behavior that determine the commitment to, and the style and proficiency of, an organization’s health and safety management” (AHRQ, 2004).<sup>24</sup> AHRQ has a survey tool to quantitatively assess 12 dimensions of patient safety culture in various health care settings (eg, hospitals, ambulatory outpatient clinics). Our institution routinely administers this survey every 18 months. Survey results from our clinic for 2009, 2011, and 2013 were compared to examine changes in our safety culture that were contemporaneous with our event reporting and formal improvement program initiatives noted previously. The number of people participating in event reporting and CQI activities is also reported because this might reflect the safety culture.

## Data analysis

Data on good catches and A3s were analyzed using descriptive statistics. Positive responses on the culture of patient safety from the AHRQ survey were compared between 2009 (baseline) and 2013 in all 12 dimensions using analysis of variance tests.

## Results

### Event learning program

Of the 560 reported events (none causing clinically meaningful harm to patients), the origin and the ending could be ascribed to a specific step on our process map for 367 of the events (Table 1). The remaining 193 events originated outside the analyzed process steps (eg, in other

**Table 1** Numbers of events generated, caught, and propagated through our processes and the resultant estimated utility of safety barriers

| Process step                                      | No. of events propagated from prior step(s) | No. of new events generated | Total events presented to this step | No. of events caught | Utility of safety barriers in this step (%) <sup>a</sup> |
|---------------------------------------------------|---------------------------------------------|-----------------------------|-------------------------------------|----------------------|----------------------------------------------------------|
| Time of consultation and presimulation directives |                                             | 133                         | 133                                 | 58                   | 44                                                       |
| Treatment planning                                | 75                                          | 100                         | 175                                 | 49                   | 28                                                       |
| Plan approval                                     | 126                                         | 16                          | 142                                 | 42                   | 30                                                       |
| Plan transfer                                     | 100                                         | 46                          | 146                                 | 18                   | 12                                                       |
| Quality assurance checks                          | 128                                         | 46                          | 174                                 | 93                   | 53                                                       |
| Treatment checks                                  | 81                                          | 26                          | 107                                 | 107                  | 100 <sup>b</sup>                                         |
| - Day of initial treatment                        |                                             |                             |                                     | 52                   | 49                                                       |
| - Remaining checks                                |                                             |                             |                                     | 55                   | 51                                                       |
|                                                   |                                             | Total = 367                 |                                     | Total = 367          |                                                          |

<sup>a</sup> Ratio of those events caught to the total number presented; eg, 58 of 133 = 44%. The utility of the safety barriers is likely overstated since there are likely errors that we are not aware of that were not caught.

<sup>b</sup> The 100% rate is artificial as it reflects the “ultimate” rate of detecting events throughout all treatment checks throughout the entire course of treatment. As shown, only 49% (52/107) were detected on the first day of therapy and the balance were found later during therapy. The term “remaining checks” refers to a wide variety of checks, including all treatment checks (day of treatment, weekly chart checks, status check, chart rounds, other not listed on the process map [eg, ad hoc checks]).

clinics or other processes). The genesis, propagation, and correction of events are noted in [Fig 2](#).

The ratio of the number of events *caught* at a particular step to the total number of events *presented* to that step (both caught and not caught; ie, allowed to propagate) is taken as a global measure of the utility of the safety barriers in that particular step ([Table 1](#)). The distribution of reported good catches among professional groups is shown in [Table 2](#). Approximately 64% of events appeared to be caused by performance issues, 20% by lack of standardized processes, and 16% by technological/environmental factors.

### Formal CQI program

Of 31 initiated A3s, 26 were implemented and at least partly sustained generating quantitative or qualitative impact on operations, 4 were discontinued, and 1 was not implemented (see [Tables 3A-3D](#) for summary data on A3s). Seventeen A3s were focused on process standardization issues while taking into account the performance based

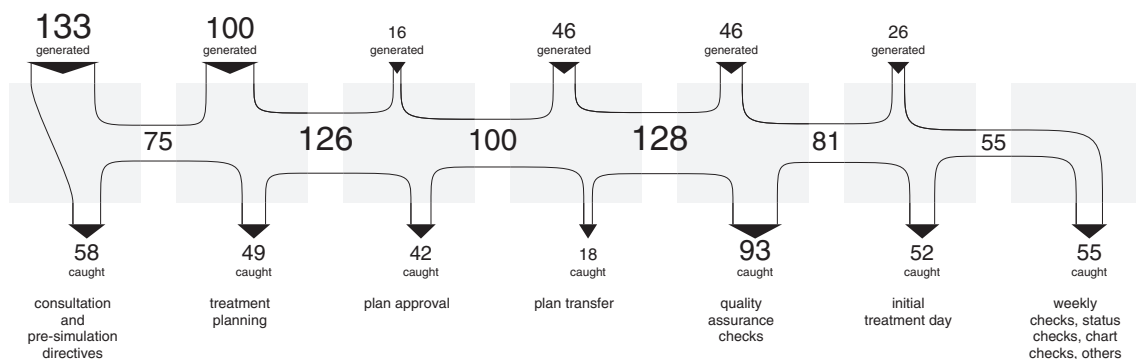
events (including 3 discontinued and 1 “canceled” A3), 8 on standardization of communications (including 1 discontinued A3), 4 on organization of spaces, and 2 on patient satisfaction. The distribution of A3s according to high-level steps in radiation therapy (RT) delivery is shown in [Table 2](#). The distribution of initiating A3s among professional staff is shown in [Table 2](#).

### Culture of patient safety

[Table 4](#) presents the AHRQ patient safety culture survey results in the department for 2009 (baseline), 2011, and 2013. Positive responses appear to have increased from 2009 to 2013 (analysis of variance;  $P < .01$ ). Further, the number of people submitting data increased from 20 in both 2001 and 2011 to 42 in 2013.

### Discussion

The results of two core components of our quality and safety program (event learning and CQI) and their potential



**Figure 2** The events flow diagram according to high-level steps in radiation therapy delivery. Each step includes the number of events caught by that step's safety barriers, the number of events newly generated, and the number of events propagating through to the next step. The width of each flow represents the number of events in that flow.

**Table 2** Distribution of good catches and A3s among professional groups

| Groups                        | No. of reported good catches (%) | No. of initiated A3s |
|-------------------------------|----------------------------------|----------------------|
| Therapists                    | 135 (24)                         | 7                    |
| Physicists                    | 133 (24)                         | 1                    |
| Dosimetrists                  | 125 (22)                         | 1                    |
| Physicians                    | 124 (22)                         | 3                    |
| Nurses                        | 34 (6)                           | 14                   |
| Administration/clerical staff | 9 (2)                            | 5                    |
| Total                         | 560                              | 31                   |

A3, a problem-solving tool that follows the formal Plan-Do-Study-Act cycle.

impact on perceptions of culture of patient safety, as quantified by the AHRQ survey, are presented. Although cause and effect cannot be established, the implementation timing of the initiatives does suggest at least some causality.

### Lessons learned from implementing our event learning program

The analysis of good catches indicates most events were generated at the time of consultation or presimulation directives and treatment planning. These results are consistent with other reports. For example, in the World Health Organization review of 7727 near misses or events leading to death or injury (incidents with adverse outcomes), 46% were associated with consultation or presimulation directives and treatment planning.<sup>25</sup> Similarly, most but not all single institution studies also note a similar distribution of events during RT process.<sup>26-37</sup> Multiple reports note a shift in where and how events are generated with the introduction of new technologies (eg, an increase in *human error* during equipment transitions when multiple machine types are used concurrently and tasks are thus less standardized).<sup>38,39</sup>

Most good catches (64%) were attributed to *performance* issues (eg, not following standardized processes, suboptimal verbal and written communications), indicating that despite the use of sophisticated technologies to help control the treatment planning process, most process steps are dependent on human behavior. This finding is in relative agreement with the World Health Organization report stating that about 60% or more of RT incidents are due to *suboptimal human performance*; New York State data indicate “failure to follow policies and procedures” in 63% of incidents, and the Radiation Oncology Safety Information System (a voluntary event registry for primarily European centers) indicate not following standards/procedures/policies in ~60% of incidents.<sup>25,40,41</sup> Indeed, this realization emphasizes the importance of having a strong culture of patient safety to help balance the

never-ending tendency of complex systems (such as RT) to drift from steady-state to the edges of chaos (or entropy).

Most events were caught during routine pretreatment QA checks, followed by checks embedded into initial treatment day. In fact, the global utility of the safety barriers incorporated into these 2 steps are relatively high (Table 1). Again, these results are not that surprising because these processes occur closest to the patient and are designed specifically to identify potential errors. Reports from other institutions indicate that pretreatment QA performed by physicists and checks/timeouts performed during the initial treatment day by therapists are particularly effective in detecting events.<sup>42</sup> On the other hand, one cannot rely solely on these end-of-the-line safety barriers because they are still imperfect; 55 events propagated past the initial treatment day (Fig 2). Thus, reducing the number of events reaching these later steps is likely still useful.

### Lessons learned from implementing our formal CQI program

The distribution of A3s is somewhat similar to the distribution of good catches (Table 2). For example, 14 A3s focused on consultation and other presimulation activities, our single largest source of good catches, and this is reassuring. The event learning and CQI programs evolved as we built our quality and safety program. Early on, the focuses for improvement events were selected by the departmental leadership (certainly with broad input from the quality and safety committee), but often not directly linked to formally reported good catches because it took some time to train people to use and become comfortable with event reporting. At later times, we did not have enough capacity to effectively respond to all reported events. Over time, we learned to appreciate how to better strategize and prioritize A3 efforts. Thus, although Table 2 suggests some degree of linkage between event reporting and CQI, it was and is not perfect.

Many of the A3 efforts were effective in spearheading and sustaining change (Tables 3A-3D). Because a relatively large portion of our events propagated through numerous steps of the radiation therapy process, approximately one-third of our A3 initiatives addressed our processes broadly. We believe that this reflects our philosophy directed toward improving systems with improvement activities focused on large portions of care delivery systems.

Nevertheless, not all A3s were successful. We needed to discontinue several A3s because of suboptimal problem definition, analysis, or implementation strategy (eg, lacking input from key stakeholders, an A3 owner leaving our institution). Indeed, others have reported similar challenges with A3-based projects and have noted that successful implementation of improvements requires close collaboration among key personnel in various functions.<sup>16,17,22</sup>

**Table 3A** A3 category: process

| A3                                     | Ownership      | Results                                                                                                   |
|----------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| Consent and creatinine levels          | Therapist      | Implemented policy to increase reliability of having a recent creatinine level at time of simulation      |
| CT simulator phone                     | Therapist      | Decreased unnecessary interruptions                                                                       |
| CyberKnife protocols                   | Therapist      | Implemented a standard simulation procedures for CyberKnife patients                                      |
| Inpatient consult requests             | Admin          | Reduced paper use and scheduler time to process requests to 5 min                                         |
| New patient orientation                | Nursing        | Developed and implemented new educational materials                                                       |
| Outpatient to inpatient protocols      | Therapist      | Implemented protocol to track patients under treatment who were admitted to the hospital                  |
| Communication of treatment information | Therapist      | Implemented assessment tool to ensure all necessary information is consistently gathered and disseminated |
| Financial counselor                    | Nursing        | Discontinued: A3 owner left facility                                                                      |
| Ordering laboratory tests              | Nursing        | Discontinued: Lack of stakeholder buy-in                                                                  |
| 3Ps checklist                          | Physician      | Checklist for pregnancy, pacemaker, and prior radiation completed for every patient before simulation     |
| Charge nurse role                      | Nursing        | Clearly defined and educated staff on what the charge nurse's role is                                     |
| Skin contours                          | Dosimetry      | Not implemented because of pending plan for a new software system                                         |
| Recovery room                          | Nursing        | Implemented standard processes to convey information regarding patients in our recovery room              |
| Financial counselor flow               | Administration | Implemented new process to ensure all patients are seen by the financial counselor                        |
| Same patient name                      | Administration | Implemented a checklist and same name alert in our electronic medical record                              |
| Medical checks change of process       | Nursing        | Decreased delays and confusion for patients being seen in clinic                                          |
| Nurse symptom management               | Nursing        | Nurses started seeing patients at high risk for unplanned admission more frequently                       |

## Lessons learned from fostering the culture of patient safety

The improvements in our safety culture survey results are reassuring. They suggest that our multifaceted quality and safety program (eg, with event learning and CQI at its core) has been at least somewhat successful. However, the pace of change is slow, with gradual improvements taking place over years. This is consistent with the experience of others.<sup>16,17,43-45</sup>

Implementing quality and safety improvement programs that require broad participation and thoughtful

input, as is the case for our good catch and A3 programs, is not fast or easy. As shown in Table 2, good catches are reported by representatives from most of the professional groups, but initiation of A3s is not well-distributed. The front-line staff (eg, nurses, radiation therapists) appears most willing to participate. Perhaps this is because they are routinely closer to the patients and thus more readily appreciate some of the consequences (eg, chaos, redoing work) of our imperfect systems, or perhaps this is because the work of physicians, physicists, and dosimetrists is perceived to be less suitable for Plan-Do-Study-Act–based improvement cycles using A3 thinking.

**Table 3B** A3 category: standardized communication

| A3                                    | Ownership      | Results                                                                 |
|---------------------------------------|----------------|-------------------------------------------------------------------------|
| CT imaging for protocol patients      | Physics        | Implemented website to improve provider/therapist communication         |
| CyberKnife phone                      | Therapist      | Implemented a dedicated phone line for therapists                       |
| Miscommunication of simulation orders | Physician      | Created new order set in MOSAIQ within the assessment tool              |
| Decrease overhead paging              | Physician      | Decreased by ~70%                                                       |
| Department phone calls                | Administration | Reduced phone calls misrouted to the department by ~85%                 |
| Late nurse communication              | Nursing        | Implemented process to communicate unexpected (>4:30 pm) nursing visits |
| Chemo-radiation coordination          | Therapist      | Discontinued: Lack of adherence to countermeasures by stakeholders      |
| Research auto-page function           | Administration | Decreased number of missed research study visits                        |

**Table 3C** A3 category: organization of spaces

| A3                            | Ownership | Results                                                                                |
|-------------------------------|-----------|----------------------------------------------------------------------------------------|
| Clean utility supply room     | Nursing   | Reduced inventory (savings of \$3,170.85 for items returned) and improved organization |
| Instrument/scope utility room | Nursing   | Improved organization                                                                  |
| Pyxis scanner swap            | Nursing   | Decreased unnecessary movement of staff                                                |
| Standardized nurse carts      | Nursing   | Implemented Kanban system                                                              |

The obstacles to broad, active, and constructive participation are likely numerous. For example, people are often understandably reluctant to report events. They do not want to admit their own errors, highlight the errors of a colleague, acknowledge the shortcomings of their “long-practiced” systems, believe that “long-broken” systems can be fixed, or think that the administration is “serious this time” about making things better. Similarly, people can have many understandable reasons not to participate in improvement initiatives (eg, competing demands on their time, the belief someone else should do it or that systems can be fixed by decree). Some may not believe that a formal improvement strategy is needed and that improvements can be readily made in an ad-hoc fashion (eg, “I saw a problem, sent few e-mails, and fixed it. We do this all the time.”). This approach certainly may be successful for “siloeed/isolated” processes; however, for many of our processes, there are upstream and downstream consequences of our work, and seemingly minor changes in one area can have negative effects elsewhere. Furthermore, because our systems are interactively complex, effects elsewhere may be unforeseen and not readily apparent. A formal structure to improvement work, with broad participation of all stakeholders, aims to better anticipate and appreciate these interactive complexities.

It has been difficult for us to motivate physicians, physicists, and dosimetrists to initiate A3s. This might be because of the perceptions of the applicability of this approach to problem-solving. This is also partially leadership’s fault because we had neither a clear strategy to motivate them nor had we developed upfront, explicit expectations. There may also be somewhat of a philosophical conflict between the traditional hierarchical view of the workplace (physicians, physicists, and dosimetrists often determine and oversee the workflow of others) versus the beliefs that systems are best managed largely by the people closest to the work. In this regard, embracing an A3 approach to CQI requires senior staff to actively engage in problem-solving as team members and relinquish some control over the systems within which they work. Others who have successfully implemented A3-based programs to CQI have noted similar challenges.<sup>16,17,46–49</sup> We need to continually address these real and understandable obstacles and evolve our strategies to foster participation because optimal improvement work requires broad participation.

The presented research has several limitations. First, the methods used to collect and analyze good catches and A3s were imperfect. There are certainly biases in event reporting (ie, not all events are reported). Thus, exact quantification of system performance and utility of safety barriers cannot be known. The global utility score as calculated is somewhat biased itself because it assumes that all RT process steps are equally designed to catch all types of reported good catches. For example, this might not be true for *plan transfer*, in which only key clinical data/information is verified for accuracy. However, this does not mean that future improvement strategies for QA during plan transfer should not be broadened to include additional data/information. Furthermore, our proposed utility measure assumes equal reporting rates between professional groups, which is fairly unlikely. For example, Smith et al found that physicians are significantly less likely to report events than their colleagues because there are specific reporting “issues” that need to be addressed to encourage reporting and create a fair culture around reporting.<sup>50</sup> The members of the Quality and Safety Committee assessed where on our process maps an event was initiated and corrected and ascribed root causes, all of which might be imprecise and can be biased. Third, the AHRQ survey could be an imperfect tool with which to measure the culture of patient safety, and different staff participated in the AHRQ survey over time. Thus, our comparisons of data over time are not ideal. Nevertheless, the AHRQ survey is generally accepted and likely is a reasonable perception-based assessment of the department’s overall patient safety culture.<sup>24</sup> Fourth, there are certainly other reasonable endpoints to consider; indeed, we have also seen increases in patient satisfaction and improved financial performance. Ideally, we would want

**Table 3D** A3 category: patient satisfaction

| A3                        | Ownership | Results                                                            |
|---------------------------|-----------|--------------------------------------------------------------------|
| Rounding in patient lobby | Nursing   | Reduced delays because of overlooked patients                      |
| Skin care for patients    | Nursing   | Standardized approach to skin care for patients with breast cancer |

A3, a problem-solving tool that follows the formal Plan-Do-Study-Act cycle; CT, computed tomography.

**Table 4** Results from the Agency for Health Care Research and Quality patient safety culture survey results for the department of radiation oncology: 2009, 2011, and 2013

|                                                   | 2009 % positive responses<br>(n = 20) | 2011 % positive response<br>(n = 20) | 2013 % positive response<br>(n = 42) |
|---------------------------------------------------|---------------------------------------|--------------------------------------|--------------------------------------|
| Communication openness                            | 77                                    | 80                                   | 80                                   |
| Information exchange with other settings          | 34                                    | 83                                   | 86                                   |
| Office process and standardization                | 54                                    | 67                                   | 82                                   |
| Organizational learning                           | 82                                    | 91                                   | 96                                   |
| Overall perceptions of patient safety and quality | 71                                    | 79                                   | 92                                   |
| Leadership support                                | 76                                    | 80                                   | 87                                   |
| Patient care tracking/follow-up                   | 66                                    | 85                                   | 86                                   |
| Patient safety and quality issues                 | 51                                    | 66                                   | 90                                   |
| Staff training                                    | 73                                    | 83                                   | 84                                   |
| Teamwork                                          | 92                                    | 99                                   | 95                                   |
| Work pressure and pace                            | 39                                    | 66                                   | 57                                   |
| Total % positive                                  | 67                                    | 76                                   | 85                                   |

There is apparent improvement in positive responses over time in essentially every metric (analysis of variance;  $P < .05$ ).

to monitor patients' clinical outcomes; however, the many variables that impact patient outcomes and the wide degree of interpatient variation present challenges to measurement. Developing the culture of patient safety is particularly valuable in the complex health care environments in which errors and activities can propagate and interact in an unpredictable manner leading to unforeseen consequences. Indeed, improvements in the culture of patient safety have been linked with a reduction in adverse events.<sup>51</sup> Therefore, our focus on measuring the culture of patient safety is reasonable.

In summary, we believe that our results are encouraging. We have been able to successfully develop, implement, and sustain event learning and CQI efforts for several years. This appears to have helped yield measurable enhancements in our culture of patient safety. Through these initiatives we have tried to create a no-blame culture in which people are comfortable reporting their concerns without fear of retribution or penalty. Based on the AHRQ survey results, this appears to have been successful. We are now working with colleagues more broadly in our Cancer Center and Health System to promote similar initiatives.

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